

SynGraph: A Dynamic Graph-LLM Synthesis Framework for Sparse Streaming User Sentiment Modeling

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SynGraph proposes a novel framework that integrates Large Language Models (LLMs) with dynamic graph modeling to address severe data sparsity in streaming sentiment analysis. It operates on Continuous-Time Dynamic Graphs (CTDGs) representing user-product interactions over time, particularly in e-commerce review platforms.

Sparsity-Aware User Categorization (With Examples)

SynGraph introduces a three-way user categorization:

- **Mid-tail Users:** Intermittently active users with bursty reviewing patterns.
- **Long-tail Users:** Sparse reviewers with few direct neighbors in the graph.
- **Extreme Users:** Nearly isolated users, sparse both temporally and structurally.

Example:

- User A writes 7 reviews in January, then none — mid-tail.
- User B has 2 reviews across a year and few common products with others — long-tail.
- User C has one review and no meaningful neighbors — extreme.

Each user type triggers a tailored augmentation pipeline.

LLM-Augmented Data Synthesis (With Examples)

SynGraph uses an LLM (e.g., GPT-3.5) to generate synthetic reviews and profiles for sparsely represented users, conditioned on local/global graph context and review history.

Example – Mid-tail User:

Reviews:

- “Great plot and writing.”
- “Liked the futuristic setting.”

Prompt:

“Generate a plausible next review that aligns with this user’s taste in sci-fi literature.”

LLM Output:

“Loved the suspense and world-building in this dystopian novel!”

The system also generates product-level profiles using co-reviewed items (second-order neighbors) and fuses this into the synthesized data.

Dynamic Graph Modeling and Interpolation (With Examples)

User-review interactions are encoded in a CTDG. SynGraph interpolates synthetic reviews into the stream to ensure at least 10 total interactions per user.

Original Stream:

T1: U41 → Book A

T6: U41 → Book B

Synthetic Interpolation:

T2: U41 → Book D (synthesized)

T4: U41 → Book C (synthesized)

This stream feeds into a dynamic encoder-decoder model to predict future sentiment ratings more accurately.

Evaluation and Results

Tested on Amazon Appliances, Gift_Cards, and Magazine_Subscriptions datasets, SynGraph showed:

- Strong performance boosts over BiLSTM, BERT, CHIM, and DC-DGNN baselines.
- Best performance with full augmentation (mid-tail + long-tail + extreme).
- 26–40% RMSE reduction depending on dataset and model.

Example Output – Before vs. After Augmentation:

Without SynGraph: U12 rating on Product X → Predicted 3.1

With SynGraph augmentation: → Predicted 4.3 (true rating: 4.0)

Quality Assessment

GPT-4 was used to evaluate synthesized reviews:

- Style similarity (LSS), sentiment alignment (SS), and rating consistency (RHS) remained high.
- Vocabulary richness was lower than original data but coherent and domain-aligned.

Why It Matters for the Survey

SynGraph is the first to integrate LLM-based synthetic data generation with dynamic graph sentiment modeling. It not only addresses user cold-start and sparsity issues but also models the **temporal evolution** of preferences. By leveraging local/global graph structure and natural language synthesis, it enriches the sparse nodes with meaningful representations—making it a foundational work in LLM-GNN hybrid systems for dynamic environments.